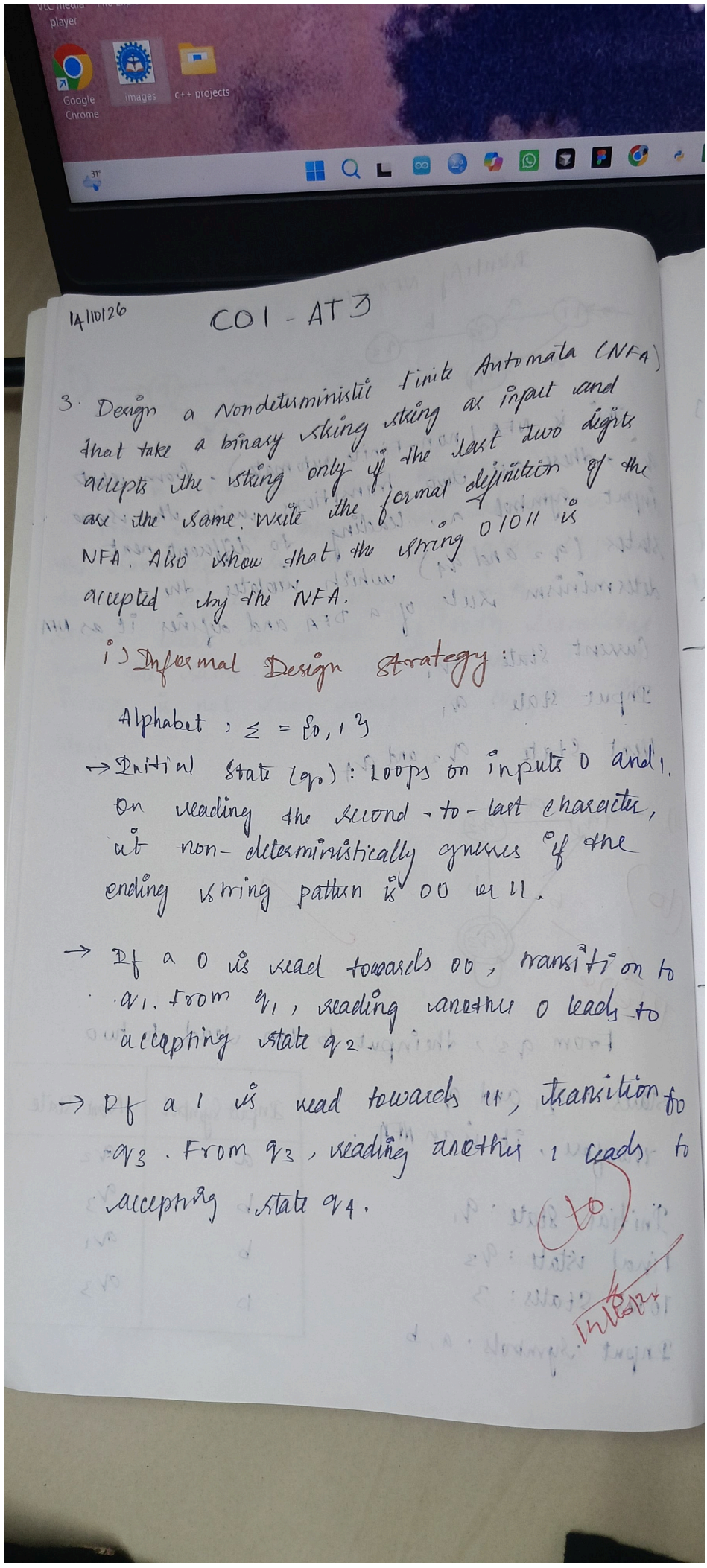


14/10/26 CO1 - AT3

3. Design a Nondeterministic Finite Automata (NFA) that take a binary string string as input and accepts the string only if the last two digits are the same. Write the formal definition of the NFA. Also show that the string 01011 is accepted by the NFA.

i) Informal Design Strategy:

Alphabet : $\Sigma = \{0, 1\}$

→ Initial state (q_0): Loops on inputs 0 and 1. On reading the second-to-last character, it non-deterministically guesses if the ending string pattern is 00 or 11.

→ If a 0 is read towards 00, transition to q_1. From q_1, reading another 0 leads to accepting state q_2.

→ If a 1 is read towards 11, transition to q_3. From q_3, reading another 1 leads to accepting state q_4.

ii) Formal Definition of the NFA:

The NFA is defined as a 5-tuple $N = (Q, \Sigma, \delta, q_0, F)$

States: $Q = \{q_0, q_1, q_2, q_3, q_4\}$

Alphabet: $\Sigma = \{0, 1\}$

Initial State: q_0

Final (Accepting) States: $F = \{q_2, q_4\}$

Transition Function (δ):

Current State	Input 0	Input 1
→ q_0	$\{q_0, q_1\}$	$\{q_0, q_3\}$
q_1	$\{q_2\}$	$\emptyset$
* q_2	$\emptyset$	$\emptyset$
q_3	$\emptyset$	$\{q_4\}$
* q_4	$\emptyset$	$\emptyset$

iii) Proof of Acceptance for string '01011'

→ Start at q_0: Active set = $\{q_0\}$

→ Read 0: $\delta(q_0, 0) = \{q_0, q_1\}$

→ Read 1:

$\delta(q_0, 1) \cup \delta(q_1, 1) = \{q_0, q_3\} \cup \emptyset = \{q_0, q_3\}$

→ Read 0:

$\delta(q_0, 0) \cup \delta(q_3, 0) = \{q_0, q_1\} \cup \emptyset = \{q_0, q_1\}$

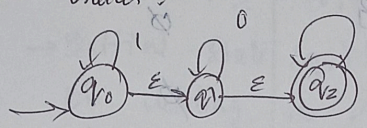
→ Read 1: $\delta(q_0, 1) \cup \delta(q_1, 1) = \{q_0, q_3\} \cup \emptyset = \{q_0, q_3\}$

→ Read 1 : $\delta(q_0, 1) \cup \delta(q_3, 1) = \{q_0, q_3\} \cup \{q_4\} = \{q_0, q_3, q_4\}$

Since the final set of active state $\{q_0, q_3, q_4\}$ contains state $q_4 \in F$, the string 01011 is accepted.

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5. A NFA with ε-transitions is given below. Find ε-closure for each state and design an equivalent DFA which accepts the same language. Draw its transition table and mark the initial & final states.



i) ε-closure of each State

The ε-closure is the set of states reachable from a state using zero or ε-transitions:

→ ε-closure $(q_0) = \{q_0, q_1, q_2\}$

→ ε-closure $(q_1) = \{q_1, q_2\}$

→ ε-closure $(q_2) = \{q_2\}$

ii) Subset Construction to DFA

Initial state:

$A = \varepsilon\text{-closure}(q_0) = \{q_0, q_1, q_2\}$

Transitions from state $A = \{q_0, q_1, q_2\}$:

On Input 0:

$\delta(q_0, 0) = \emptyset$, $\delta(q_1, 0) = \{q_1\}$, $\delta(q_2, 0) = \{q_2\}$

$\varepsilon\text{-closure}(\{q_1, q_2\}) = \{q_1, q_2\} \Rightarrow B$

On Input 1:

$\delta(q_0, 1) = \{q_0\}$, $\delta(q_1, 1) = \emptyset$, $\delta(q_2, 1) = \{q_2\}$

$\varepsilon\text{-closure}(\{q_0, q_2\}) = \{q_0, q_1, q_2\} \Rightarrow A$

Transition from state $B = \{q_1, q_2\}$:

On Input 0:

$\delta(q_1, 0) = \{q_1\}$, $\delta(q_2, 0) = \{q_2\}$

$\varepsilon\text{-closure}(\{q_1, q_2\}) = \{q_1, q_2\} \Rightarrow B$

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On Input 1:

$\delta(q_1, 1) = \emptyset$, $\delta(q_2, 1) = \{q_2\}$

$\varepsilon\text{-closure}(\{q_2\}) = \{q_2\} \Rightarrow C$

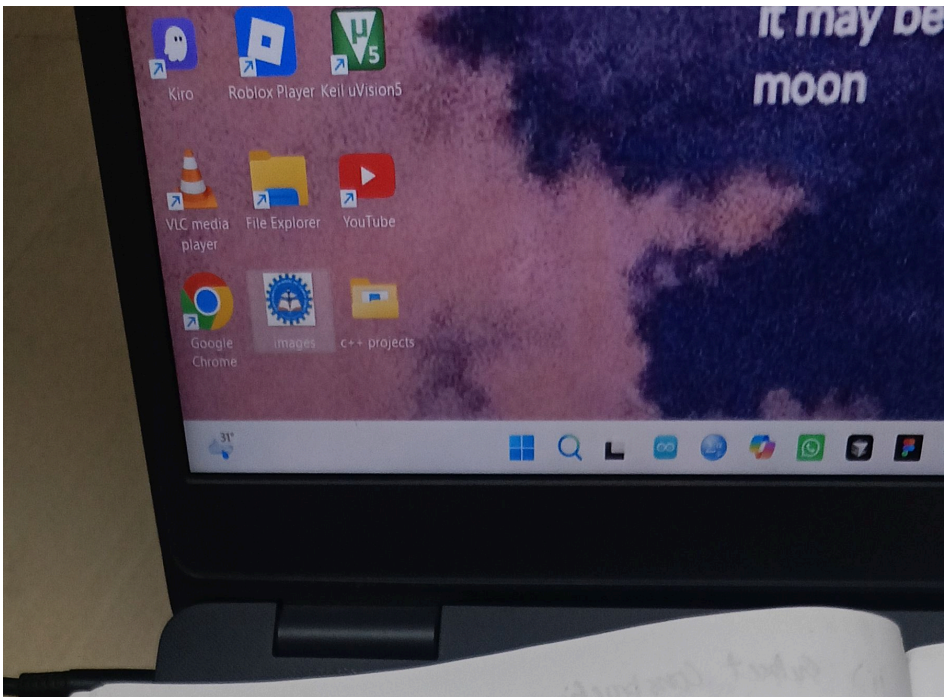
Transition from state $C = \{q_2\}$:

On Input 0:

$\delta(q_2, 0) = \{q_2\} \Rightarrow \varepsilon\text{-closure}(\{q_2\}) = C$

On Input 1:

$\delta(q_2, 1) = \{q_2\} \Rightarrow \varepsilon\text{-closure}(\{q_2\}) = C$



iii) DFA Transition Table:

→ Initial state : $A = \{q_0, q_1, q_2\}$

→ Final states : Any set containing q_2

DFA State	NFA Subset	Input 0	Input 1	State Type
→ *A	$\{q_0, q_1, q_2\}$	B	A	Initial & Final
*B	$\{q_1, q_2\}$	B	C	Final
*C	$\{q_2\}$	C	C	Final